

COMPUTER SCIENCE**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 12 printed pages.
4. It is divided into two parts: Part I and Part II.
5. It has eleven questions in all.
6. Part I is compulsory and has two questions.
7. While attempting Multiple Choice Questions in Part I, you are required to write only ONE option as the answer.
8. Part II is divided into three sections: A, B and C.
9. Each section in Part II has three questions. Any two questions have to be attempted from each section.
10. The intended marks for questions are given in brackets [].

Instruction to Supervising Examiner

1. Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

PART I – 20 MARKS*Answer all questions.**While answering questions in this Part, briefly indicate your working and reasoning, wherever required.***Question 1**

- (i) According to the Principle of Duality, the Boolean equation $A \cdot (A' + B) \cdot 1 = A \cdot B$ will be equivalent to: (*Application*) [1]
- (a) $A + (A' \cdot B) + 0 = A + B$
 - (b) $A + (A' \cdot B) + 1 = A + B$
 - (c) $A' + (A \cdot B') + 0 = A' + B'$
 - (d) $A \cdot (A' + B) + 0 = A \cdot B$

- (ii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]

Assertion: $\sim(P \wedge Q) \equiv \sim P \vee \sim Q$

Reason: This equivalence follows from the distributive law.

(Analysis)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (iii) The complement of the Boolean expression $X \cdot Y' + X' \cdot Z$ is: (Application) [1]
(a) $(X' + Y) \cdot (X + Z')$
(b) $(X + Y') \cdot (X' + Z)$
(c) $X' \cdot Y + X \cdot Z'$
(d) $(X' + Y) + (X + Z')$
- (iv) The reduced expression of $F(A, B, C) = \Sigma(4, 5, 6, 7)$ is: (Application) [1]
(a) A
(b) A'
(c) B · C
(d) 1
- (v) The canonical SOP form of $F(P, Q, R) = \Sigma(2, 5, 7)$ is: (Application) [1]
(a) $P'QR' + PQ'R + PQR$
(b) $(P + Q' + R) \cdot (P' + Q + R') \cdot (P' + Q' + R')$
(c) $P'QR + PQ'R' + PQR'$
(d) $PQ'R + P'QR + P'Q'R'$
- (vi) Consider the following constructor of class Point, where x and y are its integer data members: [1]

```
Point(int x, int y)
{
    this.x = x;
    this.y = y;
}
```

Which of the following statements are valid?

- I. **this** refers to the current object, i.e. the object being initialised by the constructor.
II. Without **this**, the statement $x = x;$ would assign the parameter x to itself, leaving the data member unchanged.
III. The keyword **this** can also be used inside a static method of the class.

(Application)

- (a) I and II
(b) II and III
(c) I and III
(d) Only III

- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]
- Assertion:** A class that implements an interface must define all the abstract methods of the interface, or else the class must itself be declared abstract.
- Reason:** The methods declared in an interface, other than default, static and private ones, are implicitly public and abstract.
- (Analysis)
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Both Assertion and Reason are false.
- (viii) The output of a 2-input NAND gate is connected to both inputs of a second NAND gate. The combination behaves as: (Analysis) [1]
- (a) an OR gate
- (b) an AND gate
- (c) a NOR gate
- (d) an XOR gate
- (ix) Write the minterm and the maxterm when $A = 0$, $B = 1$, $C = 1$ and $D = 0$. (Application) [1]
- (x) The running time of an algorithm is $T(n) = 5n^3 + 100n^2 + 3000$. State its dominant term and its complexity in Big O notation. (Analysis) [1]

Question 2

- (i) Convert the following infix notation to postfix form. [2]
- $(P + Q / R) * (S - T) + U \wedge V$ (Create)
- (ii) A matrix $N[-2 \dots 8, 3 \dots 10]$ is stored in the memory in column major wise, each element requiring 2 bytes of storage. If the address of $N[4][7]$ is 5200, find the base address (the address of $N[-2][3]$). (Application) [2]
- (iii) The following function test() is a part of some class. Assume lo and hi are the lower and upper index of the array a[]. Answer the questions given below along with dry run / working.

```
boolean test(int a[], int lo, int hi)
{
    if (lo >= hi)
        return true;
    if (a[lo] != a[hi])
        return false;
    return test(a, lo + 1, hi - 1);
}
```

- (a) What will the function test() return for $a[] = \{4, 9, 2, 9, 4\}$ with $lo = 0$, $hi = 4$, and for $a[] = \{3, 5, 5, 8\}$ with $lo = 0$, $hi = 3$? (Analysis) [2]
- (b) What is the function test() performing apart from recursion? (Analysis) [1]

- (iv) The following function `insertionSort()` is a part of some class. It sorts the array `a[ ]` in ascending order using the insertion sort technique. There are some places in the code marked by ?1?, ?2?, ?3? which may be replaced by a statement / expression so that the function works properly.

```
void insertionSort(int a[])
{
    for (int i = 1; i < a.length; i++)
    {
        int key = a[i];
        int j = ?1?;
        while (j >= 0 && ?2?)
        {
            a[j + 1] = a[j];
            j--;
        }
        ?3? = key;
    }
}
```

- (a) What is the expression or statement at ?1? (*Analysis*) [1]
(b) What is the expression or statement at ?2? (*Analysis*) [1]
(c) What is the expression or statement at ?3? (*Analysis*) [1]

PART II – 50 MARKS

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION A

Answer any two questions.

Question 3

- (i) A software company shortlists applicants for an interview if any one of the following conditions is satisfied: [5]

- The applicant holds an engineering degree and has at least two years of work experience.
OR
- The applicant does not hold an engineering degree, but has at least two years of work experience and has cleared the online test.
OR
- The applicant holds an engineering degree and has cleared the online test and is referred by an employee, but does not have two years of work experience.

The inputs are:

INPUTS	
D	holds an engineering degree
E	has at least two years of work experience
C	has cleared the online test
R	is referred by an employee

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: X – denotes the applicant is shortlisted [1 indicates YES and 0 indicates NO in all cases]

Draw the truth table for the inputs and outputs given above and write the POS expression for $X(D, E, C, R)$.

(Application)

- (ii) Reduce the above expression $X(D, E, C, R)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). Draw the logic gate diagram for the reduced expression using NOR gates only. Assume that the variables and their complements are available as inputs. (Create) [5]

Question 4

- (i)
- (a) Reduce the Boolean function $F(A, B, C, D) = \Sigma(0, 1, 4, 5, 6, 7, 13, 15)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). (Application) [4]
- (b) Draw the logic gate diagram for the reduced expression using NAND gates only. Assume that the variables and their complements are available as inputs. (Create) [1]
- (ii) What is a half adder? Draw its truth table, write the Boolean expressions for SUM and CARRY, and draw its logic circuit. (Understanding) [3]

- (iii) Using a truth table, state whether the following proposition is a tautology, a contradiction or a contingency: [2]

$$(P \Rightarrow Q) \vee (Q \Rightarrow P) \text{ (Application)}$$

Question 5

- (i)
- (a) What is a multiplexer? *(Recall)* [1]
- (b) Draw the logic circuit of a 4 : 1 multiplexer using basic gates and write its output expression. [3]
(Create)
- (c) State one application of a multiplexer. *(Understanding)* [1]
- (ii) Using the laws of Boolean algebra, prove that: [3]
$$X \cdot Y + X' \cdot Z + Y \cdot Z = X \cdot Y + X' \cdot Z$$

State the law used at each step. *(Application)*
- (iii) Draw the logic circuit of a 2-input OR gate using NAND gates only. Show the expression at each step. *(Create)* [2]

SECTION B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem. This can be achieved by using mnemonic names and comments in the program.

(Flowcharts and Algorithms are not required.)

The programs must be written in Java.

Question 6

A class **BinDec** has been defined to convert a binary number into its decimal equivalent using the recursive technique. [10]

Example: INPUT: 101101

OUTPUT: Decimal equivalent = 45 ($1 \times 32 + 0 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$)

INPUT: 1021 OUTPUT: NOT A BINARY NUMBER

Some of the members of the class are given below:

Class name : BinDec

Data members/instance variables:

bin : to store a binary number (long integer)

dec : to store its decimal equivalent

Methods / Member functions:

BinDec() : default constructor to initialise the data members to 0

void readBin() : to accept a binary number in bin

boolean isValid() : returns true if every digit of bin is 0 or 1, otherwise returns false

int convert(long b) : returns the decimal equivalent of the binary number b using the **recursive technique**

void show() : invokes convert() to store the result in dec and displays bin and dec

Specify the class BinDec giving details of the constructor(), void readBin(), boolean isValid(), int convert(long) and void show(). Define the main() function to create an object and call the functions accordingly to enable the task.

(Create)

Question 7

A class **Frame** contains a matrix of integers. The matrix of one object is to be rotated 90° clockwise, and the result compared with the matrix of another object.

[10]

Example: Matrix A (2×3) and its rotation R (3×2):

1	2	3		4	1
4	5	6	→	5	2
				6	3

The details of the members of the class are given below:

Class name : Frame

Data members/instance variables:

mat[][] : to store the integer matrix

m : number of rows

n : number of columns

Member functions/methods:

Frame(int mm, int nn) : parameterised constructor to initialise m = mm, n = nn and create the matrix

void readArray() : to accept the elements of the matrix

Frame rotate() : returns an object whose matrix is the matrix of the current object rotated 90° clockwise

boolean check(Frame P) : returns true if the matrix of the current object and that of object P are of the same order and have the same elements, otherwise returns false

void print() : to display the matrix

Specify the class Frame giving details of the constructor(int, int), void readArray(), Frame rotate(), boolean check(Frame) and void print(). Define a main() function to create two objects A and B, obtain the rotation of A, display it, and print whether B is the rotation of A. (Create)

Question 8

A class **Encrypt** has been defined to encode a sentence by replacing every vowel with the next vowel in the cycle $A \rightarrow E \rightarrow I \rightarrow O \rightarrow U \rightarrow A$, keeping its case. All other characters are left unchanged. The class also counts the vowels and consonants in the original sentence. [10]

Example: INPUT: Computer Science is FUN.

OUTPUT: Cumpatir Scoinci os FAN.

Vowels: 8 Consonants: 12

Some of the members of the class are given below:

Class name : Encrypt

Data members/instance variables:

sent : to store the sentence

newsent : to store the encoded sentence

vow, con : to store the number of vowels and consonants

Methods / Member functions:

Encrypt() : default constructor to initialise the data members with legal initial values

void accept() : to accept a sentence

char nextVowel(char ch) : returns the next vowel in the cycle if ch is a vowel (in the same case), otherwise returns ch unchanged

void encode() : forms newsent by invoking nextVowel() for every character of sent, and counts the vowels and consonants of sent

void display() : displays the original sentence, the encoded sentence and the two counts

Specify the class Encrypt giving details of the constructor(), void accept(), char nextVowel(char), void encode() and void display(). Define the main() function to create an object and call the functions accordingly to enable the task.

(Create)

SECTION C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise. This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms.

The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.

(Flowcharts are not required.)

Question 9

A pharmacy issues numbered tokens to its customers. The token numbers are kept in an array used in a circular manner: after the last position of the array, the next token is stored at the first position if it is free. Tokens are added at the rear and served from the front.

Define a class Counter with the following details:

Class name : Counter

Data members/instance variables:

tok[] : array to hold the token numbers

cap : stores the maximum capacity of the array

front : to point to the index of the front end

rear : to point to the index of the rear end

Methods / Member functions:

Counter(int max) : constructor to initialise cap = max, front = rear = 0 and to create the integer array

void push(int v) : to add a token at the rear if possible, otherwise display the message "COUNTER IS FULL"

int pop() : to remove and return the token at the front; if the array is empty, return -999

- (i) Specify the class Counter giving details of the functions void push(int) and int pop(). Assume that the other functions have been defined. The main() function and algorithm need NOT be written. (Create) [4]
- (ii) Name the entity described above and state one advantage it has over a linear queue. (Understanding) [1]

Question 10

An interface **Taxable** and a super class **Employee** have been defined. A class **IncomeTax** is to be derived from **Employee** which also implements **Taxable**, to compute the income tax of an employee from the table below:

[5]

Annual salary (Rs)	Tax
Up to 5,00,000	Nil
5,00,001 to 10,00,000	10% of the amount above 5,00,000
Above 10,00,000	50,000 + 20% of the amount above 10,00,000

A health cess of CESS % of the tax is added to the tax.

The details of the interface and the classes are given below:

Interface name : Taxable
double CESS : constant, the health cess rate (4.0)
double computeTax() : to compute and return the tax
Class name : Employee

Data members/instance variables:

name, pan : to store the name and PAN of the employee
salary : to store the annual salary (in decimals)

Methods / Member functions:

Employee(...) : parameterised constructor to assign values to the data members
void display() : to display the employee details
Class name : IncomeTax

Data members/instance variables:

tax : to store the income tax (in decimals)

Methods / Member functions:

IncomeTax(...) : parameterised constructor to assign values to the data members of both the classes
double computeTax() : computes the tax as per the table, adds the cess, stores it in tax and returns it
void display() : displays the employee details along with the tax

Assume that the interface **Taxable** and the super class **Employee** have been defined. Using the concept of inheritance, specify the class **IncomeTax** giving the details of the constructor(...), double computeTax() and void display().

The interface, super class, main function and algorithm need NOT be written.

(Create)

Question 11

- (i) A linked list is formed from the objects of the class Node. The class structure of the Node is given below:

[2]

```
class Node
{
    int data;
    Node next;
}
```

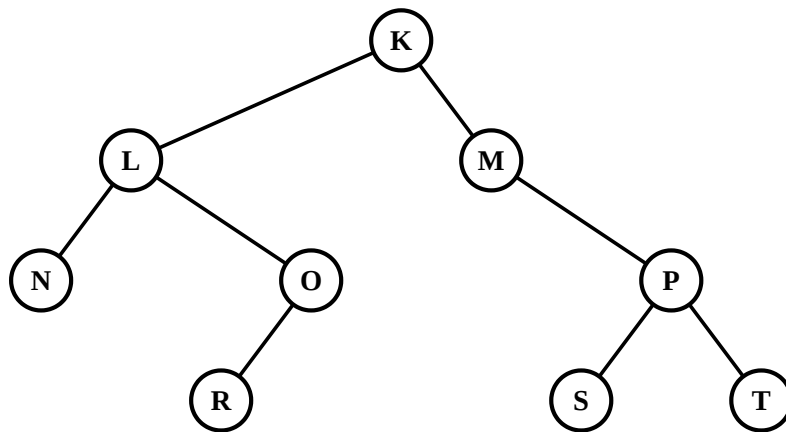
Write an Algorithm OR a Method to delete the first node whose data is k from an existing linked list and return the reference to the first node of the resulting list.

The method declaration is as follows:

Node delete(Node start, int k)

(Analysis)

- (ii) Answer the following questions from the diagram of a Binary Tree given below:



- (a) Write the pre-order traversal of the above tree structure. (Understanding)
- (b) State the size of the left subtree and the height of the tree. (Understanding)
- (c) Name the parent of node R and the sibling of node S. (Understanding)

[1]

[1]

[1]

— — — —